

DELAWARE COASTAL, DE, USA

WMO: 724093

Lat: 38.690N

Lon: 75.363W

Elev: 14

StdP: 101.16

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 13764

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.1	-7.1	-18.7	0.7	-6.9	-16.2	0.9	-5.0	11.7	10.4	9.7	10.3	2.4	310	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	33.9	24.2	32.6	23.8	31.3	23.2	25.8	31.0	25.2	30.1	24.6	29.0	4.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.4	19.4	27.9	23.9	18.7	27.3	23.1	17.9	26.7	79.8	31.3	76.8	30.1	74.4	29.0	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	8.2	7.3	-12.5	36.2	3.0	1.6	-14.6	37.4	-16.4	38.3	-18.0	39.1	-20.2	40.3		
			-13.3	27.1	2.8	0.6	-15.4	27.6	-17.0	27.9	-18.6	28.3	-20.7	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.1	2.5	3.7	7.4	13.1	18.1	22.9	25.5	24.7	21.2	15.2	9.5	5.0
	DBStd	9.21	5.84	5.24	5.33	4.88	4.36	3.49	2.87	2.65	3.43	4.55	4.68	5.14
	HDD10.0	813	243	186	116	21	1	0	0	0	0	7	66	173
	HDD18.3	2288	492	409	339	169	60	5	0	0	12	119	268	414
	CDD10.0	2317	10	11	37	115	252	387	479	455	336	169	49	17
	CDD18.3	751	0	0	2	14	53	142	221	197	98	22	2	1
	CDH23.3	6580	1	1	17	133	477	1277	2160	1700	669	139	6	0
	CDH26.7	2401	0	0	2	34	145	476	911	622	184	28	0	0
Wind	WSAvg	3.1	3.2	3.4	3.7	3.8	3.3	3.0	2.7	2.5	2.8	2.8	2.8	2.9
Precipitation	PrecAvg	1170	81	76	93	87	92	98	117	126	107	85	81	95
	PrecMax	1551	130	156	181	141	182	275	188	243	226	194	151	189
	PrecMin	937	33	21	9	39	26	23	42	16	9	1	9	44
	PrecStd	183	28	32	43	26	45	65	40	61	59	55	43	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.2	21.3	25.7	30.1	32.2	34.7	36.3	34.8	33.0	29.4	23.8	21.3
		MCWB	16.3	15.7	16.9	19.0	21.8	24.0	25.4	25.1	23.7	21.9	18.1	17.8
	2%	DB	17.8	18.0	22.2	27.3	30.3	32.8	34.2	32.9	30.7	27.0	21.7	18.7
		MCWB	15.1	13.2	15.3	18.0	21.4	23.5	24.4	24.5	22.8	20.7	16.7	16.3
	5%	DB	15.0	15.1	19.0	24.0	28.0	31.2	32.9	31.7	28.7	24.5	19.2	16.3
		MCWB	12.2	11.0	13.7	16.8	20.5	23.0	23.9	23.8	22.0	19.4	15.8	13.9
	10%	DB	11.9	12.3	16.4	21.8	26.2	29.5	31.4	30.1	27.2	22.5	17.3	13.5
		MCWB	9.1	8.7	11.9	15.6	19.6	22.1	23.6	23.1	21.4	18.3	13.9	11.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.5	17.2	18.2	21.2	23.5	25.5	26.9	26.7	25.7	23.9	19.7	18.7
		MCDB	19.4	20.0	23.6	26.7	29.3	31.7	32.8	31.8	29.7	27.0	21.9	20.4
	2%	WB	15.5	14.4	16.6	19.6	22.4	24.7	26.0	25.8	24.5	22.4	18.1	16.6
		MCDB	17.7	17.0	20.5	25.2	28.1	30.3	31.6	30.6	27.8	25.5	20.3	18.3
	5%	WB	12.6	12.1	14.8	18.1	21.5	24.1	25.4	25.1	23.8	20.9	16.5	14.3
		MCDB	14.7	14.3	18.6	22.5	26.8	29.1	30.5	29.5	26.7	23.5	18.8	16.0
	10%	WB	9.1	8.9	12.5	16.7	20.5	23.4	24.7	24.4	22.9	19.2	14.6	11.6
		MCDB	11.3	12.0	15.6	20.9	25.0	27.9	29.4	28.4	25.6	21.7	16.8	13.1
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.7	10.2	11.2	10.6	10.0	9.8	9.5	9.7	10.4	10.3	9.3
		MCDBR	12.4	13.4	14.1	14.3	13.0	11.8	11.3	10.8	11.4	11.7	12.1	11.7
		MCWBR	10.2	10.0	8.5	7.4	6.0	4.6	4.0	4.1	5.0	6.3	8.0	10.2
	5% WB	MCDBR	11.9	12.2	12.6	12.2	11.7	10.2	9.8	9.3	8.6	9.6	10.1	11.1
		MCWBR	10.7	10.7	9.2	7.5	5.7	4.5	3.9	4.0	4.4	6.1	8.1	10.5
Clear-Sky Solar Irradiance	taub	0.316	0.320	0.358	0.394	0.443	0.488	0.508	0.503	0.412	0.360	0.327	0.315	
	taud	2.479	2.468	2.399	2.318	2.212	2.134	2.095	2.094	2.325	2.472	2.516	2.522	
	Ebn at Noon	861	908	901	884	842	803	782	775	832	849	842	838	
	Edh at Noon	80	92	109	125	142	154	159	155	116	91	77	72	
All-Sky Solar Radiation	RadAvg	2.13	2.97	3.98	5.09	5.75	6.29	6.14	5.48	4.48	3.35	2.41	1.87	
	RadStd	0.19	0.31	0.33	0.37	0.64	0.48	0.30	0.35	0.45	0.34	0.22	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.66	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (69 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+139	+82

Nomenclature: See separate page